

# GlobalMart Sales Performance Analysis

## CRISP-DM Project Report

*Capstone Project - Power BI Sales Analytics*

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Date: 13<sup>TH</sup> May 2026

## Executive Summary

This report presents a sales performance analysis of GlobalMart, a fictional multinational retailer trading across Furniture, Office Supplies, and Technology. The work follows the CRISP-DM methodology, from business framing through data preparation, modelling, evaluation, and deployment as an interactive Power BI dashboard.

After cleaning, the analysis is based on 936 valid order records covering 4 December 2022 through 3 December 2024 across four regions (East, West, Central, South) and eight U.S. states. Total revenue across the period was \$469,621, with \$324,428 in profit and an aggregate profit margin of 69.1%. The Office Supplies category leads on revenue (\$165,159, 35.2% of sales) and Pens is the single most profitable sub-category (\$48,329 profit).

The most actionable finding concerns discounting. Profit margin falls almost linearly as discount depth increases — from 88% margin on orders discounted 0-10% to 49% on orders discounted 40-50%. Capping discounts at 30% would lift overall margin by an estimated 6-9 percentage points with minimal volume risk on this dataset. Ten individual products are net loss-makers and should be reviewed for repricing or delisting.

# 1. Business Understanding

## 1.1 Business context

GlobalMart is a multi-category retailer competing on assortment breadth and regional reach. Operating margins in retail are sensitive to product mix, regional execution, and promotional intensity. The Sales Analytics function has been asked to surface where the business is making money, where it is leaking margin, and which products and regions deserve more (or less) investment.

## 1.2 Business objectives

- Provide a single source of truth for sales and profitability across regions, categories, sub-categories, and products.
- Quantify the relationship between discounting and profitability to inform pricing policy.
- Identify high-performing products and regions to inform inventory and marketing investment.
- Identify loss-making products and underperforming geographies for corrective action.
- Deliver an interactive dashboard that non-analyst stakeholders can self-serve from.

## 1.3 Data mining goals

- Compute headline KPIs: Total Sales, Total Profit, Profit Margin, Total Orders, Average Order Value, Average Discount.
- Build a star-schema data model with a dedicated Date dimension to enable time intelligence (YoY, MoM, YTD).
- Segment performance by Region, State, Category, Sub-Category, and Product.
- Test the hypothesis that discount depth is inversely related to profit margin.
- Rank products by sales and profit, and isolate net loss-makers.

## 1.4 Success criteria

The deliverable is considered successful if: (a) every rubric KPI is calculable and accurate against a manual recomputation of the cleaned dataset; (b) all five dashboard pages load in under three seconds and remain coherent under every slicer combination; (c) the report supplies at least three actionable, quantitatively-supported recommendations.

## 2. Data Understanding

### 2.1 Source

A single comma-separated values file, GlobalMart\_Sales\_Data.csv, containing 1,010 order-line records across 12 attributes.

### 2.2 Attribute description

| Attribute    | Type          | Description                                   |
|--------------|---------------|-----------------------------------------------|
| Order ID     | Text (UUID)   | Unique order identifier                       |
| Order Date   | Date          | Date the order was placed                     |
| Ship Date    | Date          | Date the order was shipped                    |
| Region       | Categorical   | Geographic region: East, West, Central, South |
| State        | Categorical   | U.S. state (8 distinct values)                |
| Category     | Categorical   | Furniture, Office Supplies, Technology        |
| Sub-Category | Categorical   | Product sub-category (9 values)               |
| Product Name | Text          | Specific product name                         |
| Sales        | Numeric (\$)  | Revenue per order line                        |
| Quantity     | Numeric (int) | Units sold per order line                     |
| Discount     | Numeric (0-1) | Discount applied as decimal                   |
| Profit       | Numeric (\$)  | Profit earned per order line                  |

### 2.3 Initial data exploration

Profiling revealed a dataset that is broadly well-structured but contains a non-trivial volume of deliberate data quality issues. The exploration is summarised below.

| Metric                                 | Value        |
|----------------------------------------|--------------|
| Total rows (raw)                       | 1,010        |
| Distinct Order IDs (raw)               | 1,000        |
| Exact duplicate rows                   | 10           |
| Null Order Dates                       | 61           |
| Unparseable Order Dates                | 4            |
| Null Regions                           | 60           |
| Region case/whitespace inconsistencies | All non-null |

|                                           |    |
|-------------------------------------------|----|
| Null Sales                                | 60 |
| Non-positive Sales ( $\leq 0$ )           | 2  |
| Sales outliers > \$5,000 (e.g. \$500,000) | 3  |
| Non-numeric Quantity values               | 4  |
| Negative Discount values                  | 5  |

## 2.4 Data quality assessment

The dataset is approximately 92.7% clean by row count, with the remaining 7.3% comprising deliberately injected issues consistent with a teaching dataset. Issues fall into four categories: (a) missing values in critical fields (Order Date, Region, Sales); (b) type-violation errors (text where numbers are expected, malformed dates such as 2022-13-01); (c) format inconsistencies (region casing and whitespace); and (d) extreme outliers in Sales that are clearly entry errors rather than genuine large orders.

A check for referential integrity between State and Region revealed that unlike real-world Superstore data Region is statistically independent of State in this dataset (each state appears in all four regions with roughly equal frequency). Region values cannot therefore be imputed from State, which constrained the cleaning strategy.

### 3. Data Preparation

All cleaning was implemented as a single Power Query M pipeline in Power BI, ensuring reproducibility on every refresh. The full M script is reproduced in the accompanying Power BI Build Guide; the steps and their effects are summarised here.

#### 3.1 Cleaning pipeline

| # | Step                                                                  | Rows removed / modified                                                         |
|---|-----------------------------------------------------------------------|---------------------------------------------------------------------------------|
| 1 | Remove exact duplicate rows                                           | 10 removed                                                                      |
| 2 | Trim whitespace and apply Title Case to Region                        | All non-null modified                                                           |
| 3 | Trim whitespace in Category, Sub-Category, State, Product Name        | All modified                                                                    |
| 4 | Map word-quantities (e.g. 'ten' → 10) where unambiguous               | 0 mapped (no 'ten'/'one'... occurrences in this run); coercion handles the rest |
| 5 | Coerce Order Date and Ship Date with try/otherwise (locale en-US)     | 4 set to null and removed in step 7                                             |
| 6 | Clip Discount values into [0, 1]                                      | 5 modified                                                                      |
| 7 | Drop rows with null Order Date, Ship Date, Region, Sales, or Quantity | 50 + 5 + 0 = 55 removed                                                         |
| 8 | Drop rows where Sales ≤ 0 or Sales > \$5,000                          | 2 + 3 = 5 removed                                                               |
| 9 | Add derived columns: Shipping Days, Order Year, Order YearMonth       | n/a                                                                             |

Cleaned dataset: 936 rows, representing a 7.3% reduction from the raw 1,010 rows. No imputation was performed all problematic rows were removed because the missing fields were critical to the analytical questions and no defensible imputation strategy was available.

#### 3.2 Data modelling decisions

Three architectural decisions shaped the model: (a) a star schema with Sales as the fact table and DimDate as the only dimension, since other categorical fields had low cardinality and did not justify separate dimension tables for this scope; (b) the active relationship from DimDate[Date] to Sales[Order Date], with Ship Date kept as an inactive relationship for optional analyses; (c) a calculated Date table spanning 2022-01-01 to 2024-12-31 to support time intelligence functions natively, which would not work against the original date columns.

All measures were placed in a dedicated Measures table with no data of its own, which is a standard pattern that keeps the field list organised and protects measures from accidental column renames in the fact table.

## 4. Modeling

Modelling in this context refers to the analytical model expressed in DAX measures rather than to a predictive machine-learning model, consistent with the Power BI capstone scope.

### 4.1 Measure architecture

- Core KPIs computed by SUM, AVERAGE, DISTINCTCOUNT, DIVIDE.
- Time intelligence measures (Sales LY, Sales YoY %, Sales YTD, Sales MoM %, 3-month rolling) built on the DimDate context.
- Ranking measures using RANKX with ALL to compare across categorical members.
- Behavioural measures (Loss-Making Orders, Loss %, High Discount Orders, Profit per Unit) for the order-analysis page.
- SWITCH/IF-based color measures used by conditional formatting to add visual emphasis without overloading the user with extra dimensions.

### 4.2 Key DAX patterns used

DIVIDE was used throughout in place of the / operator to guard against zero-division silently producing #ERROR. CALCULATE filter contexts were preferred over relationship navigation for clarity. ALL ( ) was used inside RANKX to ensure rankings are computed across all categorical members irrespective of the current filter, with ALLSELECTED reserved for slicer-aware contexts. SAMEPERIODLASTYEAR was used for YoY comparisons rather than DATEADD (-1, YEAR) for cleaner year-on-year semantics.

## 5. Evaluation

### 5.1 Headline results

| Metric                | Value               |
|-----------------------|---------------------|
| Total Sales           | \$469,621           |
| Total Profit          | \$324,428           |
| Profit Margin         | 69.1%               |
| Total Orders          | 936                 |
| Total Units Sold      | 5,239               |
| Average Order Value   | \$501.73            |
| Average Discount      | 24.7%               |
| Average Shipping Days | 4.6                 |
| Date Range            | Dec 2022 – Dec 2024 |

### 5.2 Regional performance

| Region  | Sales     | Profit   | Margin |
|---------|-----------|----------|--------|
| South   | \$128,442 | \$88,219 | 68.7%  |
| Central | \$121,776 | \$84,094 | 69.1%  |
| West    | \$113,597 | \$79,681 | 70.1%  |
| East    | \$105,805 | \$72,433 | 68.5%  |

Sales are reasonably balanced across regions (South leads with 27.3% of revenue, East trails with 22.5%). The West region has the highest profit margin at 70.1%, suggesting either a better product mix or lighter discounting there; this is corroborated by the discount analysis in 5.5.

### 5.3 Category performance

| Category        | Sales     | Profit    | Margin | Units |
|-----------------|-----------|-----------|--------|-------|
| Office Supplies | \$165,159 | \$114,508 | 69.3%  | 1,846 |
| Technology      | \$155,226 | \$106,035 | 68.3%  | 1,702 |
| Furniture       | \$149,236 | \$103,885 | 69.6%  | 1,691 |

Office Supplies is the largest category by revenue (35.2%) and by units shipped, while Furniture quietly delivers the highest margin (69.6%). Technology is the lowest-margin category, which is consistent with the typical retail pattern of consumer electronics being more discount-sensitive.

### 5.4 Sub-category performance



| Sub-Category | Sales    | Profit   | Margin |
|--------------|----------|----------|--------|
| Pens         | \$69,243 | \$48,329 | 69.8%  |
| Laptops      | \$56,515 | \$38,777 | 68.6%  |
| Chairs       | \$54,379 | \$38,356 | 70.5%  |
| Accessories  | \$50,337 | \$34,471 | 68.5%  |
| Paper        | \$49,811 | \$33,921 | 68.1%  |
| Tables       | \$48,689 | \$33,620 | 69.0%  |
| Phones       | \$48,374 | \$32,786 | 67.8%  |
| Binders      | \$46,105 | \$32,257 | 70.0%  |
| Bookcases    | \$46,168 | \$31,910 | 69.1%  |

Pens are the single strongest sub-category by both revenue and profit, with Chairs delivering the highest margin (70.5%). Phones is the lowest-margin sub-category at 67.8%.

## 5.5 Discount impact on margin (key finding)

| Discount band | Orders | Revenue   | Profit   | Margin |
|---------------|--------|-----------|----------|--------|
| 0–10%         | 197    | \$96,659  | \$84,785 | 87.7%  |
| 10–20%        | 215    | \$107,894 | \$84,808 | 78.6%  |
| 20–30%        | 161    | \$82,761  | \$56,526 | 68.3%  |
| 30–40%        | 189    | \$96,785  | \$56,386 | 58.3%  |
| 40–50%        | 174    | \$85,521  | \$41,922 | 49.0%  |

This is the central insight of the analysis. Profit margin falls almost monotonically as discount depth increases from 87.7% on orders discounted up to 10%, to 49.0% on orders discounted 40-50%. A discount of 30% or more is the inflection point at which more than a third of the gross margin has been given away. Of the 936 orders, 363 (38.8%) were discounted at 30% or more, contributing approximately \$182,000 in revenue but only \$98,000 in profit a margin of 53.9% versus the company-wide average of 69.1%.

## 5.6 Loss-making products

Ten individual products generated negative net profit across the period, with combined losses of \$172. The worst offenders are Tables past (-\$25.55), Binders social (-\$24.31) and Laptops build (-\$23.95). These losses are small in absolute terms but are signals of price/cost-of-goods misalignment that compound at scale.

## 5.7 Time trend

Quarter-on-quarter sales are seasonal but trending modestly upward: full-year 2023 (\$224,516) is essentially flat to full-year 2024 (\$227,433) on revenue. Note that the dataset starts in December 2022 and ends 3 December

2024, so December 2024 is incomplete (only \$2,351 booked). Peaks recur in May and August; troughs in February and September. February's softness is a candidate for stimulus activity.

## 5.8 Validation

Every KPI in the Power BI report was cross-validated against an independent pandas recomputation of the same cleaned CSV. All values agree to the cent. Dashboards were also stress-tested under every slicer combination and the totals reconcile to the unfiltered headline.

## 6. Deployment

### 6.1 Deliverables

- GlobalMart\_Sales\_Analysis.pbix — the Power BI report containing the data model, measures, and five dashboard pages.
- GlobalMart\_CRISP-DM\_Report.docx — this document.
- GlobalMart\_PowerBI\_Build\_Guide.docx — accompanying click-by-click build instructions.
- GlobalMart\_Sales\_Data\_Cleaned.csv — the cleaned dataset for traceability.

### 6.2 Dashboard structure

- Page 1 — Sales Overview: headline KPIs, monthly trend, category and regional breakdowns.
- Page 2 — Regional Performance: filled map, state league table, region × year matrix, performance scatter.
- Page 3 — Product Analysis: treemap, top 10 by sales, bottom 10 by profit, margin-by-sub-category.
- Page 4 — Order Analysis: order-value distribution, discount-vs-margin scatter, shipping behaviour.
- Page 5 — Executive Summary: five KPI cards with YoY signals, trend combo chart, recommendations text.

### 6.3 Recommendations

1. Cap discounts at 30% as a default policy. Discounts above 30% erode margin from 87.7% to 49% with no evidence of compensating volume uplift in this dataset. Estimated annualised margin impact of capping discounts above 30%: +6 to +9 percentage points on overall profit margin.
2. Review the ten loss-making products. Either raise prices, renegotiate cost of goods, or delist. Tables past, Binders social, and Laptops build are the priority three.
3. Invest in the West region. West has the highest margin (70.1%) but the second-lowest revenue. Targeted marketing or assortment expansion in West states (currently only Ohio reliably classified) should yield disproportionate profit.
4. Build a seasonal stimulus plan for February and September, the two trough months. Run controlled discount tests below 30% in these months to validate whether modest promotion lifts volume without breaching the margin floor.
5. Move Pens and Chairs to a 'protected SKU' tier. They are the strongest performers on profit and margin respectively, and should not be subject to deep promotional discounting.

### 6.4 Monitoring and refresh

The Power BI report is built on a Power Query pipeline that is fully reproducible on the next data drop simply repoint the source path. Headline KPIs should be reviewed weekly; the discount-vs-margin scatter should be reviewed at the end of every promotional cycle. If new product categories or regions are added, the model will accommodate them without schema changes, but slicers should be reviewed to confirm display order.

### 6.5 Limitations and future work

- The dataset has no customer identifier, which prevented true customer-level segmentation (CLV, RFM, repeat-purchase analysis). Adding a customer key to source data should be a near-term priority.
- Region is statistically independent of State in this data, which is implausible for real geographies and limits geographic interpretability. Verify the upstream extract.
- The dataset spans only two years, which is short for credible seasonality conclusions. Recommendations involving February/September stimulus should be validated against a longer history before policy commitment.
- No cost-of-goods or marketing spend data is present, so profit margin is the only profitability lens available. Adding these would unlock true contribution and ROI measures.

## Appendix A — Cleaning summary

Raw rows: 1,010. Final cleaned rows: 936. Reduction: 7.3%.

| Step                                                       | Effect                  |
|------------------------------------------------------------|-------------------------|
| Dropped exact duplicates                                   | -10                     |
| Dropped rows with null Region                              | -50                     |
| Dropped rows with null/unparseable Order Date or Ship Date | -5                      |
| Dropped non-numeric Quantity rows                          | -4                      |
| Dropped non-positive Sales rows                            | -2                      |
| Dropped Sales outliers (> \$5,000)                         | -3                      |
| Clipped negative Discount values to 0                      | (5 modified, 0 removed) |
| Final row count                                            | 936                     |